

Antonio Carmagnani

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PROFILE

Experience in computational design and data analysis for real estate and cities, spanning academic research and architecture practice. Recently completed a Master's Degree in Architecture Computation at The Royal Danish Academy, with a focus on computational and data methods for real estate and urban development. Hands-on skills span Python, QGIS, and machine learning, parametric workflows in Grasshopper, and building data tools that turn spatial and financial data into decisions. Motivated by work where the built environment, technology, and data intersect.

EXPERIENCE

Project Assistant in International Infrastructure Projects @ Sweco Danmark A/S

Copenhagen, DK

02.2026 - 07.2026

- Support consulting teams delivering internationally funded infrastructure projects in water, energy, and transport across emerging and developing economies, financed by development banks including the World Bank, EBRD, IFDK, KfW and EIB.
- Developed a pricing tool to support bid decisions, parametrizing expert fee bands and project economics for financial viability.
- Reviewed engineering designs across multiple infrastructure projects under FIDIC Yellow Book arrangements, checking specifications against subconsultants' technical drawings.

Intern in Responsible Business Conduct @ Ramboll Management Consulting

Copenhagen, DK

05.2025 - 08.2025

- Conducted human rights due diligence for corporate clients, applying the UN Guiding Principles on Business and Human Rights and OECD Guidelines to assess salient risks across global supply chains.
- Mapped the global supply chains of 20+ critical raw materials under the EU Critical Raw Materials Act, profiling producing companies, sectors, and geographies from authoritative industry databases.

Architect @ Isay Weinfeld Architects

São Paulo, BR

02.2023 - 08.2024

- Delivered high-end residential, hotel, and mixed-use projects across South America, North America, Europe, the Middle East, and Asia at one of Brazil's most prominent architecture studios.
- Developed building facades across the studio's design and technical development teams, carrying envelopes from the early compositional studies through to resolved construction build-ups on residential, hotel, and mixed-use buildings.
- Resolved facade build-ups and their materials, fixings, and jointing, working through preliminary structural sizing with the project's engineers before the detail was documented.
- Coordinated with structural and building services engineers on facade packages, reviewing their input against the design intent and resolving conflicts before documentation moved on.
- Ran daylight and solar-exposure simulations with Ladybug in Grasshopper to drive massing and orientation decisions, quantifying light incidence across building layouts.
- Built a Grasshopper plugin in Python that generates bespoke staircases from a floor plan, cutting staircase modeling from days to around 20 minutes and enabling far more design iterations and geometric complexity.

Trainee in Real Estate Product Development @ OSPA

São Paulo, BR

02.2022 - 10.2022

- Produced 15+ feasibility studies for mixed-use and residential developments across Brazil and the US (São Paulo, Rio de Janeiro, Porto Alegre, Miami) for developer clients.
- Developed volumetric massing studies in Revit, resolving circulation, unit layouts, ground-floor active frontage, and parking against zoning metrics.

Architect @ Felipe Hess Architects

São Paulo, BR

07.2021 - 02.2022

- Developed high-end residential projects in São Paulo and Rio de Janeiro from preliminary studies through technical documentation, working in Revit, AutoCAD, and SketchUp.
- Designed and detailed the facades of residential and commercial buildings, resolving materials, openings, and envelope build-up from preliminary studies through to construction documentation.
- Produced BIM documentation in Revit and construction drawings across design phases.

Intern in Project Management @ Itaú-Unibanco

São Paulo, BR

08.2020 - 07.2021

- Reviewed technical documentation from architecture and engineering firms prior to construction authorization, checking drawings against building, accessibility, and fire-safety norms (NBR standards).
- Supported technical review of new office towers under construction at the bank's São Paulo headquarters campus, evaluating design documentation for compliance and environmental performance.

Independent Consultant @ Self-employed

São Paulo, BR and Copenhagen, DK

07.2018 - Present

- Delivered independent consulting to clients across the AEC sector, spanning product and UX advice, Revit and Dynamo automation, and financial analysis tooling.
- Co-founded and ran a small practice delivering projects for 10+ clients in the AEC sector, responsible for client acquisition, scoping, pricing, and delivery.
- Advised a software startup in the built environment over two years on product direction, UI and UX, and applied AI, testing releases and returning improved Revit and Dynamo scripts.

EDUCATION

Master's Degree in Architecture Computation @ The Royal Danish Academy

Copenhagen, DK

07.2024 - 07.2026

Semester Programme in Building Technology @ Architecture Association

London, UK

07.2022 - 01.2023

*06.2019 - 07.2019**07.2018 - 12.2023***PROJECTS****Computational Deployable Structures Workshop @ Southeast University x The Royal Danish Academy****Nanjing, CN**

- Selected as one of a small group from The Royal Danish Academy for a two-week international workshop at Southeast University, applying origami-based deployable-structure principles (used in satellite deployment) to architectural structures.
- Developed parametric models exploring foldable geometry, material systems, and structural performance, working across computational design, geometry, and engineering.

Rammed Blue Biomass & Computational Optimization @ The Royal Danish Academy**Copenhagen, DK**

- Developed a new construction technique and material system using blue biomass (seashells and algae), applying LCA metrics and parametric simulations to optimize structural performance and environmental impact.
- Project was selected as a semester highlight and served as the material foundation for the school's installation at the Venice Architecture Biennale 2025, presented in collaboration with CITA, GXN, and 3XN at the Arsenale.

Venice Architecture Biennale 2025 @ The Royal Danish Academy, CITA, GXN, 3XN**Venice, IT**

- Contributed to the Matanzas panel by developing geospatial mappings to analyze local material sources and supply flows, producing cartographic work that informed the site's material research.
- Contributed to the Veneto panel by producing the physical exhibition panels displayed in the 7-metre installation at the Arsenale, which used the blue biomass composite technique developed during the semester project.

SKILLS

- building_and_facade: Facade and envelope design, Construction detailing and technical documentation, Multidisciplinary BIM coordination, Daylight and solar simulation (Ladybug)
- Digital Tools: Rhino + Grasshopper, Revit, AutoCAD, Adobe Suite (InDesign, Illustrator, Photoshop)
- Data & Analysis: Python, QGIS, SQL, Power BI, Excel / VBA, HTML
- Languages: English (fluent), Portuguese (native), Spanish (fluent), Danish (limited working proficiency), Italian (limited working proficiency)